

JESS Thermodynamic Database v8.9

Chemical species in reactions with Protactinium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
Acac-1 Acetylacetonate ion				
-1	17272-66-1	267	C(5)H(7)O(2)	99.1094
Acetic-1 Acetate ion				
-1	71-50-1	558	C(2)H(3)O(2)	59.0446
Benzoic-1 Benzoate ion; Benzenecarboxylate ion				
-1	766-76-7	208	C(7)H(5)O(2)	121.116
Br-1 Bromide ion				
-1	24959-67-9	1213	Br(1)	79.9040
Br2(1) Bromine, liquid				
0	---	285	Br(2)	159.808
Cat-2 Catecholate ion; 1,2-Dihydroxybenzoate ion; 1,2-Benzenedioate ion; Pyrocatecholate ion; Oxyphenoxide				
-2	19021-48-8	333	C(6)H(4)O(2)	108.097
Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl(1)	35.4530
Cl2(g) Chlorine gas				
0	7782-50-5	589	Cl(2)	70.9060

CO3-2 Carbonate ion				
-2	3812-32-6	1435	C(1)O(3)	60.0092
e-1 Electron				
-1	---	3399	E(1)	0.000000
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C(10)H(12)N(2)O(8)	288.214
EtDiAm Ethylenediamine; 1,2-Ethanediamine; 1,2-Diaminoethane; en !				
0	107-15-3	879	C(2)H(8)N(2)	60.0989
F-1 Fluoride ion				
-1	16984-48-8	1090	F(1)	18.9984
Gly-1 Glycinate ion; Aminoacetate ion				
-1	---	1356	C(2)H(4)N(1)O(2)	74.0593
Glycolic-1 Glycolate ion				
-1	57122-18-6	342	C(2)H(3)O(3)	75.0440
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H(1)	1.00794
H+1_234TriOHPentDioic-2				
-1	---	5	C(5)H(7)O(7)	179.106
H+1_Cl-1				
0	---	11	Cl(1)H(1)	36.4609
H+1_CO3-2				

-1	---	253	C(1)H(1)O(3)	61.0171
H+1_F-1				
0	---	142	F(1)H(1)	20.0063
H+1_IO3-1 Hydrogen iodate				
0	---	10	H(1)I(1)O(3)	175.906
H+1_PaO2+1_CO3-2				
0	---	1	C(1)H(1)O(5)Pa(1)	324.056
H+1_PaO2+1_OH-1(3)_234TriOHPentDioic-2				
-3	---	1	C(5)H(10)O(12)Pa(1)	493.167
H+1_PaO2+1_PO4-3				
-1	---	2	H(1)O(6)P(1)Pa(1)	359.018
H+1_PO4-3 Hydrogenphosphate ion				
-2	---	317	H(1)O(4)P(1)	95.9795
H+1_SO4-2 Hydrogensulfate ion				
-1	---	83	H(1)O(4)S(1)	97.0655
H+1(2)_PaO2+1_CO3-2(2)				
-1	---	1	C(2)H(2)O(8)Pa(1)	385.073
H+1(2)_PaO2+1_PO4-3				
0	---	2	H(2)O(6)P(1)Pa(1)	360.026
H+1(2)_PaO2+1_PO4-3(2)				
-3	---	2	H(2)O(10)P(2)Pa(1)	454.998
H+1(2)_PO4-3 Dihydrogenphospate ion				
-1	14066-20-7	284	H(2)O(4)P(1)	96.9875

H+1 (3) _PaO2+1 _CO3-2 (3)				
-2	---	1	C (3) H (3) O (11) Pa (1)	446.090
H+1 (3) _PaO2+1 _PO4-3 (3)				
-5	---	2	H (3) O (14) P (3) Pa (1)	550.977
H+1 (4) _PaO2+1 _CO3-2 (4)				
-3	---	1	C (4) H (4) O (14) Pa (1)	507.107
H+1 (4) _PaO2+1 _PO4-3 (2)				
-1	---	2	H (4) O (10) P (2) Pa (1)	457.014
H+1 (4) _PaO2+1 _PO4-3 (4)				
-7	---	2	H (4) O (18) P (4) Pa (1)	646.957
H+1 (5) _PaO2+1 _PO4-3 (5)				
-9	---	2	H (5) O (22) P (5) Pa (1)	742.937
H+1 (6) _PaO2+1 _PO4-3 (3)				
-2	---	2	H (6) O (14) P (3) Pa (1)	554.001
H2O Water				
0	7732-18-5	5947	H (2) O (1)	18.0153
I-1 Iodide ion				
-1	20461-54-5	894	I (1)	126.900
IO3-1 Iodate ion				
-1	15454-31-6	291	I (1) O (3)	174.898
NH3 Ammonia, aqueous				
0	---	1462	H (3) N (1)	17.0305

NO3-1 Nitrate ion				
-1	14797-55-8	573	N(1)O(3)	62.0049
NTA-3 Nitrilotriacetate ion; N,N-bis(carboxymethyl)glycinate ion; Triglycollamate ion; Triglycinate ion; NTA ion				
-3	28528-44-1	751	C(6)H(6)N(1)O(6)	188.117
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O(2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073
Oxalic-2 Oxalate ion				
-2	338-70-5	1241	C(2)O(4)	88.0196
Pa+2				
2	---	1	Pa(1)	231.040
Pa+3 Protoactinium(III) ion				
3	28971-98-4	4	Pa(1)	231.040
Pa+4 Protoactinium(IV) ion; Protactinium(IV) ion				
4	22537-59-3	162	Pa(1)	231.040
Pa+4_Acac-1				
3	---	1	C(5)H(7)O(2)Pa(1)	330.149
Pa+4_Acac-1(2)				
2	---	1	C(10)H(14)O(4)Pa(1)	429.259
Pa+4_Acetic-1				

3	---	1	C (2) H (3) O (2) Pa (1)	290.085
Pa+4_Acetic-1 (2)				
2	---	1	C (4) H (6) O (4) Pa (1)	349.129
Pa+4_Acetic-1 (3)				
1	---	1	C (6) H (9) O (6) Pa (1)	408.174
Pa+4_Acetic-1 (4)				
0	---	1	C (8) H (12) O (8) Pa (1)	467.219
Pa+4_Acetic-1 (5)				
-1	---	1	C (10) H (15) O (10) Pa (1)	526.263
Pa+4_Acetic-1 (6)				
-2	---	1	C (12) H (18) O (12) Pa (1)	585.308
Pa+4_Benzoic-1				
3	---	1	C (7) H (5) O (2) Pa (1)	352.156
Pa+4_Benzoic-1 (2)				
2	---	1	C (14) H (10) O (4) Pa (1)	473.271
Pa+4_Benzoic-1 (3)				
1	---	1	C (21) H (15) O (6) Pa (1)	594.387
Pa+4_Benzoic-1 (4)				
0	---	1	C (28) H (20) O (8) Pa (1)	715.502
Pa+4_Benzoic-1 (5)				
-1	---	1	C (35) H (25) O (10) Pa (1)	836.617
Pa+4_Benzoic-1 (6)				
-2	---	1	C (42) H (30) O (12) Pa (1)	957.733
Pa+4_Br-1				
3	---	1	Br (1) Pa (1)	310.944

Pa+4_Br-1 (2)				
2	---	1	Br (2) Pa (1)	390.848
Pa+4_Br-1 (3)				
1	---	1	Br (3) Pa (1)	470.752
Pa+4_Br-1 (4)				
0	---	1	Br (4) Pa (1)	550.656
Pa+4_Br-1 (4) (s) Protoactinium bromide; Protoactinium tetrabromide				
0	---	1	Br (4) Pa (1)	550.656
Pa+4_Br-1 (5)				
-1	---	1	Br (5) Pa (1)	630.560
Pa+4_Br-1 (6)				
-2	---	1	Br (6) Pa (1)	710.464
Pa+4_Cat-2				
2	---	1	C (6) H (4) O (2) Pa (1)	339.137
Pa+4_Cat-2 (2)				
0	---	1	C (12) H (8) O (4) Pa (1)	447.233
Pa+4_Cat-2 (3)				
-2	---	1	C (18) H (12) O (6) Pa (1)	555.330
Pa+4_Cl-1				
3	---	2	Cl (1) Pa (1)	266.493
Pa+4_Cl-1 (2)				
2	---	2	Cl (2) Pa (1)	301.946
Pa+4_Cl-1 (3)				
1	---	1	Cl (3) Pa (1)	337.399

Pa+4_Cl-1 (4)				
0	---	1	Cl (4) Pa (1)	372.852
Pa+4_Cl-1 (4) (s) Protactinium(IV) chloride				
0	---	2	Cl (4) Pa (1)	372.852
Pa+4_Cl-1 (5)				
-1	---	1	Cl (5) Pa (1)	408.305
Pa+4_Cl-1 (6)				
-2	---	1	Cl (6) Pa (1)	443.758
Pa+4_CO3-2				
2	---	1	C (1) O (3) Pa (1)	291.049
Pa+4_CO3-2 (2)				
0	---	1	C (2) O (6) Pa (1)	351.058
Pa+4_CO3-2 (2) (s) Protoactinium(IV) carbonate				
0	---	1	C (2) O (6) Pa (1)	351.058
Pa+4_CO3-2 (3)				
-2	---	1	C (3) O (9) Pa (1)	411.068
Pa+4_CO3-2 (4)				
-4	---	1	C (4) O (12) Pa (1)	471.077
Pa+4_CO3-2 (5)				
-6	---	1	C (5) O (15) Pa (1)	531.086
Pa+4_CO3-2 (6)				
-8	---	1	C (6) O (18) Pa (1)	591.095
Pa+4_EtDiAm				
4	---	1	C (2) H (8) N (2) Pa (1)	291.139

Pa+4_EtDiAm(2)				
4	---	1	C(4)H(16)N(4)Pa(1)	351.238
Pa+4_EtDiAm(3)				
4	---	1	C(6)H(24)N(6)Pa(1)	411.337
Pa+4_F-1				
3	---	3	F(1)Pa(1)	250.038
Pa+4_F-1(2)				
2	---	3	F(2)Pa(1)	269.037
Pa+4_F-1(3)				
1	---	1	F(3)Pa(1)	288.035
Pa+4_F-1(4)				
0	---	1	F(4)Pa(1)	307.034
Pa+4_F-1(5)				
-1	---	1	F(5)Pa(1)	326.032
Pa+4_F-1(6)				
-2	---	1	F(6)Pa(1)	345.030
Pa+4_Gly-1				
3	---	1	C(2)H(4)N(1)O(2)Pa(1)	305.099
Pa+4_Gly-1(2)				
2	---	1	C(4)H(8)N(2)O(4)Pa(1)	379.159
Pa+4_Gly-1(3)				
1	---	1	C(6)H(12)N(3)O(6)Pa(1)	453.218
Pa+4_Glycolic-1				
3	---	1	C(2)H(3)O(3)Pa(1)	306.084

Pa+4_Glycolic-1(2)				
2	---	1	C(4)H(6)O(6)Pa(1)	381.128
Pa+4_Glycolic-1(3)				
1	---	1	C(6)H(9)O(9)Pa(1)	456.172
Pa+4_Glycolic-1(4)				
0	---	1	C(8)H(12)O(12)Pa(1)	531.216
Pa+4_Glycolic-1(5)				
-1	---	1	C(10)H(15)O(15)Pa(1)	606.260
Pa+4_Glycolic-1(6)				
-2	---	1	C(12)H(18)O(18)Pa(1)	681.304
Pa+4_H+1_CO3-2				
3	---	1	C(1)H(1)O(3)Pa(1)	292.057
Pa+4_H+1_PO4-3				
2	---	2	H(1)O(4)P(1)Pa(1)	327.020
Pa+4_H+1(10)_PO4-3(5)				
-1	---	2	H(10)O(20)P(5)Pa(1)	715.977
Pa+4_H+1(12)_PO4-3(6)				
-2	---	2	H(12)O(24)P(6)Pa(1)	812.965
Pa+4_H+1(2)_CO3-2(2)				
2	---	1	C(2)H(2)O(6)Pa(1)	353.074
Pa+4_H+1(2)_PO4-3				
3	---	2	H(2)O(4)P(1)Pa(1)	328.028
Pa+4_H+1(2)_PO4-3(2)				
0	---	2	H(2)O(8)P(2)Pa(1)	422.999

Pa+4_H+1 (3)_CO3-2 (3)				
1	---	1	C (3) H (3) O (9) Pa (1)	414.091
Pa+4_H+1 (3)_PO4-3 (3)				
-2	---	2	H (3) O (12) P (3) Pa (1)	518.979
Pa+4_H+1 (4)_CO3-2 (4)				
0	---	1	C (4) H (4) O (12) Pa (1)	475.109
Pa+4_H+1 (4)_PO4-3 (2)				
2	---	2	H (4) O (8) P (2) Pa (1)	425.015
Pa+4_H+1 (4)_PO4-3 (4)				
-4	---	2	H (4) O (16) P (4) Pa (1)	614.958
Pa+4_H+1 (5)_CO3-2 (5)				
-1	---	1	C (5) H (5) O (15) Pa (1)	536.126
Pa+4_H+1 (5)_PO4-3 (5)				
-6	---	2	H (5) O (20) P (5) Pa (1)	710.938
Pa+4_H+1 (6)_CO3-2 (6)				
-2	---	1	C (6) H (6) O (18) Pa (1)	597.143
Pa+4_H+1 (6)_PO4-3 (3)				
1	---	2	H (6) O (12) P (3) Pa (1)	522.002
Pa+4_H+1 (6)_PO4-3 (6)				
-8	---	2	H (6) O (24) P (6) Pa (1)	806.917
Pa+4_H+1 (8)_PO4-3 (4)				
0	---	2	H (8) O (16) P (4) Pa (1)	618.990
Pa+4_I-1				
3	---	1	I (1) Pa (1)	357.940

Pa+4_I-1 (2)				
2	---	1	I (2) Pa (1)	484.840
Pa+4_I-1 (3)				
1	---	1	I (3) Pa (1)	611.740
Pa+4_I-1 (4)				
0	---	1	I (4) Pa (1)	738.640
Pa+4_I-1 (5)				
-1	---	1	I (5) Pa (1)	865.540
Pa+4_I-1 (6)				
-2	---	1	I (6) Pa (1)	992.440
Pa+4_IO3-1				
3	---	1	I (1) O (3) Pa (1)	405.938
Pa+4_IO3-1 (2)				
2	---	1	I (2) O (6) Pa (1)	580.836
Pa+4_IO3-1 (3)				
1	---	1	I (3) O (9) Pa (1)	755.735
Pa+4_IO3-1 (4)				
0	---	1	I (4) O (12) Pa (1)	930.633
Pa+4_IO3-1 (5)				
-1	---	1	I (5) O (15) Pa (1)	1105.53
Pa+4_IO3-1 (6)				
-2	---	1	I (6) O (18) Pa (1)	1280.43
Pa+4_NH3				
4	---	1	H (3) N (1) Pa (1)	248.071

Pa+4_NH3 (2)				
4	---	1	H (6) N (2) Pa (1)	265.101
Pa+4_NH3 (3)				
4	---	1	H (9) N (3) Pa (1)	282.132
Pa+4_NH3 (4)				
4	---	1	H (12) N (4) Pa (1)	299.162
Pa+4_NH3 (5)				
4	---	1	H (15) N (5) Pa (1)	316.193
Pa+4_NH3 (6)				
4	---	1	H (18) N (6) Pa (1)	333.223
Pa+4_NO3-1				
3	---	1	N (1) O (3) Pa (1)	293.045
Pa+4_NO3-1 (2)				
2	---	1	N (2) O (6) Pa (1)	355.050
Pa+4_NO3-1 (3)				
1	---	1	N (3) O (9) Pa (1)	417.055
Pa+4_NO3-1 (4)				
0	---	1	N (4) O (12) Pa (1)	479.060
Pa+4_NO3-1 (5)				
-1	---	1	N (5) O (15) Pa (1)	541.065
Pa+4_NO3-1 (6)				
-2	---	1	N (6) O (18) Pa (1)	603.069
Pa+4_OH-1				
3	---	2	H (1) O (1) Pa (1)	248.047

Pa+4 _OH-1 _SO4-2				
1	---	1	H (1) O (5) Pa (1) S (1)	344.105
Pa+4 _OH-1 (2)				
2	---	3	H (2) O (2) Pa (1)	265.055
Pa+4 _OH-1 (2) _SO4-2				
0	---	1	H (2) O (6) Pa (1) S (1)	361.112
Pa+4 _OH-1 (3)				
1	---	2	H (3) O (3) Pa (1)	282.062
Pa+4 _OH-1 (4)				
0	---	3	H (4) O (4) Pa (1)	299.069
Pa+4 _OH-1 (4) _ (s) Protoactinium(IV) hydroxide				
0	---	1	H (4) O (4) Pa (1)	299.069
Pa+4 _OH-1 (5)				
-1	---	1	H (5) O (5) Pa (1)	316.077
Pa+4 _OH-1 (6)				
-2	---	1	H (6) O (6) Pa (1)	333.084
Pa+4 _Oxalic-2				
2	---	2	C (2) O (4) Pa (1)	319.060
Pa+4 _Oxalic-2 (2)				
0	---	2	C (4) O (8) Pa (1)	407.079
Pa+4 _Oxalic-2 (3)				
-2	---	1	C (6) O (12) Pa (1)	495.099
Pa+4 _Oxalic-2 (4)				
-4	---	1	C (8) O (16) Pa (1)	583.118

Pa+4_Phenol-1				
3	---	1	C (6) H (5) O (1) Pa (1)	324.145
Pa+4_Phenol-1 (2)				
2	---	1	C (12) H (10) O (2) Pa (1)	417.250
Pa+4_Phenol-1 (3)				
1	---	1	C (18) H (15) O (3) Pa (1)	510.355
Pa+4_Phenol-1 (4)				
0	---	1	C (24) H (20) O (4) Pa (1)	603.460
Pa+4_Phenol-1 (5)				
-1	---	1	C (30) H (25) O (5) Pa (1)	696.566
Pa+4_Phenol-1 (6)				
-2	---	1	C (36) H (30) O (6) Pa (1)	789.671
Pa+4_SCN-1				
3	---	1	C (1) N (1) Pa (1) S (1)	289.118
Pa+4_SCN-1 (2)				
2	---	1	C (2) N (2) Pa (1) S (2)	347.195
Pa+4_SCN-1 (3)				
1	---	1	C (3) N (3) Pa (1) S (3)	405.273
Pa+4_SCN-1 (4)				
0	---	1	C (4) N (4) Pa (1) S (4)	463.351
Pa+4_SCN-1 (5)				
-1	---	1	C (5) N (5) Pa (1) S (5)	521.429
Pa+4_SCN-1 (6)				
-2	---	1	C (6) N (6) Pa (1) S (6)	579.506

Pa+4 _SO4-2				
2	---	2	O (4) Pa (1) S (1)	327.098
Pa+4 _SO4-2 (2)				
0	---	2	O (8) Pa (1) S (2)	423.155
Pa+4 _SO4-2 (3)				
-2	---	1	O (12) Pa (1) S (3)	519.213
Pa+4 _SO4-2 (4)				
-4	---	1	O (16) Pa (1) S (4)	615.270
Pa+4 _SO4-2 (5)				
-6	---	1	O (20) Pa (1) S (5)	711.328
Pa+4 _SO4-2 (6)				
-8	---	1	O (24) Pa (1) S (6)	807.386
Pa+4 (4) _OH-1 (12)				
4	---	1	H (12) O (12) Pa (4)	1128.25
Pa+4 (6) _OH-1 (15)				
9	---	1	H (15) O (15) Pa (6)	1641.35
Pa+5 Protoactinium(V) ion				
5	---	30	Pa (1)	231.040
Pa+5 _Cl-1				
4	---	2	Cl (1) Pa (1)	266.493
Pa+5 _Cl-1 _OH-1 (3)				
1	---	1	Cl (1) H (3) O (3) Pa (1)	317.515
Pa+5 _Cl-1 (2)				
3	---	1	Cl (2) Pa (1)	301.946

Pa+5_F-1				
4	---	4	F (1) Pa (1)	250.038
Pa+5_F-1 (2)				
3	---	4	F (2) Pa (1)	269.037
Pa+5_F-1 (3)				
2	---	3	F (3) Pa (1)	288.035
Pa+5_F-1 (4)				
1	---	2	F (4) Pa (1)	307.034
Pa+5_F-1 (5)				
0	---	2	F (5) Pa (1)	326.032
Pa+5_F-1 (6)				
-1	---	2	F (6) Pa (1)	345.030
Pa+5_F-1 (7)				
-2	---	2	F (7) Pa (1)	364.029
Pa+5_F-1 (8)				
-3	---	1	F (8) Pa (1)	383.027
Pa+5_IO3-1				
4	---	2	I (1) O (3) Pa (1)	405.938
Pa+5_IO3-1 (2)				
3	---	1	I (2) O (6) Pa (1)	580.836
Pa+5_NO3-1				
4	---	2	N (1) O (3) Pa (1)	293.045
Pa+5_NO3-1 (2)				
3	---	2	N (2) O (6) Pa (1)	355.050

Pa+5_NO3-1 (4)				
1	---	1	N (4) O (12) Pa (1)	479.060
Pa+5_OH-1 (2)				
3	---	1	H (2) O (2) Pa (1)	265.055
Pa+5_OH-1 (2)_EDTA-4				
-1	---	1	C (10) H (14) N (2) O (10) Pa (1)	553.269
Pa+5_OH-1 (3)				
2	---	2	H (3) O (3) Pa (1)	282.062
Pa+5_OH-1 (4)				
1	---	2	H (4) O (4) Pa (1)	299.069
Pa+5_OH-1 (5)				
0	---	2	H (5) O (5) Pa (1)	316.077
Pa+5_Oxalic-2				
3	---	2	C (2) O (4) Pa (1)	319.060
Pa+5_Oxalic-2 (2)				
1	---	2	C (4) O (8) Pa (1)	407.079
Pa+5_Oxalic-2 (3)				
-1	---	1	C (6) O (12) Pa (1)	495.099
Pa+5_SO4-2				
3	---	2	O (4) Pa (1) S (1)	327.098
Pa+5_SO4-2 (2)				
1	---	1	O (8) Pa (1) S (2)	423.155
Pa (s) Protoactinium				
0	7440-13-3	16	Pa (1)	231.040

Pa₂O₅ (s)				
Protoactinium(V) oxide				
0	---	5	O (5) Pa (2)	542.077
PaO+3				
3	---	15	O (1) Pa (1)	247.039
PaO+3_Cl-1_OH-1				
1	---	1	Cl (1) H (1) O (2) Pa (1)	299.500
PaO+3_F-1				
2	---	1	F (1) O (1) Pa (1)	266.038
PaO+3_NTA-3				
0	---	1	C (6) H (6) N (1) O (7) Pa (1)	435.156
PaO+3_NTA-3 (2)				
-3	---	1	C (12) H (12) N (2) O (13) Pa (1)	623.273
PaO+3_OH-1				
2	---	11	H (1) O (2) Pa (1)	264.047
PaO+3_OH-1 (2)				
1	---	2	H (2) O (3) Pa (1)	281.054
PaO+3_OH-1 (4)				
-1	---	2	H (4) O (5) Pa (1)	315.069
PaO+3_SO4-2				
1	---	2	O (5) Pa (1) S (1)	343.097
PaO+3_SO4-2 (2)				
-1	---	2	O (9) Pa (1) S (2)	439.155
PaO+3_SO4-2 (3)				
-3	---	1	O (13) Pa (1) S (3)	535.212

PaO2+1				
1	---	108	O (2) Pa (1)	263.039
PaO2+1_Acetic-1				
0	---	1	C (2) H (3) O (4) Pa (1)	322.083
PaO2+1_Acetic-1 (2)				
-1	---	1	C (4) H (6) O (6) Pa (1)	381.128
PaO2+1_Acetic-1 (3)				
-2	---	1	C (6) H (9) O (8) Pa (1)	440.173
PaO2+1_Acetic-1 (4)				
-3	---	1	C (8) H (12) O (10) Pa (1)	499.217
PaO2+1_Benzoic-1				
0	---	1	C (7) H (5) O (4) Pa (1)	384.154
PaO2+1_Benzoic-1 (2)				
-1	---	1	C (14) H (10) O (6) Pa (1)	505.270
PaO2+1_Benzoic-1 (3)				
-2	---	1	C (21) H (15) O (8) Pa (1)	626.385
PaO2+1_Benzoic-1 (4)				
-3	---	1	C (28) H (20) O (10) Pa (1)	747.501
PaO2+1_Benzoic-1 (5)				
-4	---	1	C (35) H (25) O (12) Pa (1)	868.616
PaO2+1_Br-1				
0	---	1	Br (1) O (2) Pa (1)	342.943
PaO2+1_Br-1 (2)				
-1	---	1	Br (2) O (2) Pa (1)	422.847

PaO₂+1_Br-1 (3)				
-2	---	1	Br (3) O (2) Pa (1)	502.751
PaO₂+1_Br-1 (4)				
-3	---	1	Br (4) O (2) Pa (1)	582.655
PaO₂+1_Br-1 (5)				
-4	---	1	Br (5) O (2) Pa (1)	662.559
PaO₂+1_Cat-2				
-1	---	1	C (6) H (4) O (4) Pa (1)	371.135
PaO₂+1_Cat-2 (2)				
-3	---	1	C (12) H (8) O (6) Pa (1)	479.232
PaO₂+1_Cl-1				
0	---	1	Cl (1) O (2) Pa (1)	298.492
PaO₂+1_Cl-1 (2)				
-1	---	1	Cl (2) O (2) Pa (1)	333.945
PaO₂+1_Cl-1 (3)				
-2	---	1	Cl (3) O (2) Pa (1)	369.398
PaO₂+1_Cl-1 (4)				
-3	---	1	Cl (4) O (2) Pa (1)	404.851
PaO₂+1_Cl-1 (5)				
-4	---	1	Cl (5) O (2) Pa (1)	440.304
PaO₂+1_CO3-2				
-1	---	1	C (1) O (5) Pa (1)	323.048
PaO₂+1_CO3-2 (2)				
-3	---	1	C (2) O (8) Pa (1)	383.057

PaO₂+1_CO₃-2 (3)				
-5	---	1	C (3) O (11) Pa (1)	443.066
PaO₂+1_CO₃-2 (4)				
-7	---	1	C (4) O (14) Pa (1)	503.076
PaO₂+1_CO₃-2 (5)				
-9	---	1	C (5) O (17) Pa (1)	563.085
PaO₂+1_EtDiAm				
1	---	1	C (2) H (8) N (2) O (2) Pa (1)	323.138
PaO₂+1_EtDiAm (2)				
1	---	1	C (4) H (16) N (4) O (2) Pa (1)	383.237
PaO₂+1_F-1				
0	---	1	F (1) O (2) Pa (1)	282.037
PaO₂+1_F-1 (2)				
-1	---	1	F (2) O (2) Pa (1)	301.036
PaO₂+1_F-1 (3)				
-2	---	1	F (3) O (2) Pa (1)	320.034
PaO₂+1_F-1 (4)				
-3	---	1	F (4) O (2) Pa (1)	339.032
PaO₂+1_F-1 (5)				
-4	---	1	F (5) O (2) Pa (1)	358.031
PaO₂+1_Gly-1				
0	---	1	C (2) H (4) N (1) O (4) Pa (1)	337.098
PaO₂+1_Gly-1 (2)				
-1	---	1	C (4) H (8) N (2) O (6) Pa (1)	411.157

PaO2+1_Glycolic-1				
0	---	1	C (2) H (3) O (5) Pa (1)	338.083
PaO2+1_Glycolic-1 (2)				
-1	---	1	C (4) H (6) O (8) Pa (1)	413.127
PaO2+1_Glycolic-1 (3)				
-2	---	1	C (6) H (9) O (11) Pa (1)	488.171
PaO2+1_Glycolic-1 (4)				
-3	---	1	C (8) H (12) O (14) Pa (1)	563.215
PaO2+1_I-1				
0	---	1	I (1) O (2) Pa (1)	389.939
PaO2+1_I-1 (2)				
-1	---	1	I (2) O (2) Pa (1)	516.839
PaO2+1_I-1 (3)				
-2	---	1	I (3) O (2) Pa (1)	643.739
PaO2+1_I-1 (4)				
-3	---	1	I (4) O (2) Pa (1)	770.639
PaO2+1_I-1 (5)				
-4	---	1	I (5) O (2) Pa (1)	897.539
PaO2+1_IO3-1				
0	---	1	I (1) O (5) Pa (1)	437.937
PaO2+1_IO3-1 (2)				
-1	---	1	I (2) O (8) Pa (1)	612.835
PaO2+1_IO3-1 (3)				
-2	---	1	I (3) O (11) Pa (1)	787.733

PaO₂+1_I₂O₃-1 (4)				
-3	---	1	I (4) O (14) Pa (1)	962.632
PaO₂+1_I₂O₃-1 (5)				
-4	---	1	I (5) O (17) Pa (1)	1137.53
PaO₂+1_NH₃				
1	---	1	H (3) N (1) O (2) Pa (1)	280.069
PaO₂+1_NH₃ (2)				
1	---	1	H (6) N (2) O (2) Pa (1)	297.100
PaO₂+1_NH₃ (3)				
1	---	1	H (9) N (3) O (2) Pa (1)	314.130
PaO₂+1_NH₃ (4)				
1	---	1	H (12) N (4) O (2) Pa (1)	331.161
PaO₂+1_NH₃ (5)				
1	---	1	H (15) N (5) O (2) Pa (1)	348.191
PaO₂+1_NO₃-1				
0	---	1	N (1) O (5) Pa (1)	325.044
PaO₂+1_NO₃-1 (2)				
-1	---	1	N (2) O (8) Pa (1)	387.049
PaO₂+1_NO₃-1 (3)				
-2	---	1	N (3) O (11) Pa (1)	449.054
PaO₂+1_NO₃-1 (4)				
-3	---	1	N (4) O (14) Pa (1)	511.058
PaO₂+1_NO₃-1 (5)				
-4	---	1	N (5) O (17) Pa (1)	573.063

PaO2+1_OH-1				
0	---	1	H (1) O (3) Pa (1)	280.046
PaO2+1_OH-1_(am.,s)				
0	---	4	H (1) O (3) Pa (1)	280.046
PaO2+1_OH-1(2)				
-1	---	1	H (2) O (4) Pa (1)	297.054
PaO2+1_OH-1(3)				
-2	---	2	H (3) O (5) Pa (1)	314.061
PaO2+1_OH-1(4)				
-3	---	1	H (4) O (6) Pa (1)	331.068
PaO2+1_OH-1(5)				
-4	---	1	H (5) O (7) Pa (1)	348.076
PaO2+1_Phenol-1				
0	---	1	C (6) H (5) O (3) Pa (1)	356.144
PaO2+1_Phenol-1(2)				
-1	---	1	C (12) H (10) O (4) Pa (1)	449.249
PaO2+1_Phenol-1(3)				
-2	---	1	C (18) H (15) O (5) Pa (1)	542.354
PaO2+1_Phenol-1(4)				
-3	---	1	C (24) H (20) O (6) Pa (1)	635.459
PaO2+1_Phenol-1(5)				
-4	---	1	C (30) H (25) O (7) Pa (1)	728.564
PaO2+1_SCN-1				
0	---	1	C (1) N (1) O (2) Pa (1) S (1)	321.117

PaO2+1_SCN-1 (2)				
-1	---	1	C (2) N (2) O (2) Pa (1) S (2)	379.194
PaO2+1_SCN-1 (3)				
-2	---	1	C (3) N (3) O (2) Pa (1) S (3)	437.272
PaO2+1_SCN-1 (4)				
-3	---	1	C (4) N (4) O (2) Pa (1) S (4)	495.350
PaO2+1_SCN-1 (5)				
-4	---	1	C (5) N (5) O (2) Pa (1) S (5)	553.427
PaO2+1_SO4-2				
-1	---	1	O (6) Pa (1) S (1)	359.096
PaO2+1_SO4-2 (2)				
-3	---	1	O (10) Pa (1) S (2)	455.154
PaO2+1_SO4-2 (3)				
-5	---	1	O (14) Pa (1) S (3)	551.212
PaO2+1_SO4-2 (4)				
-7	---	1	O (18) Pa (1) S (4)	647.269
PaO2+1 (2)_CO3-2_(s)				
0	---	1	C (1) O (7) Pa (2)	586.087
PaO2 (am. , s) Protactinium dioxide				
0	---	3	O (2) Pa (1)	263.039
PaO2 (s) Protoactinium dioxide; Protoactinium(IV) oxide				
0	12036-03-2	8	O (2) Pa (1)	263.039
Phenol-1 Phenolate ion; Hydroxybenzoate ion				

-1	3229-70-7	154	C(6)H(5)O(1)	93.1051
PO4-3 Phosphate ion; Phosphate ion				
-3	14265-44-2	1594	O(4)P(1)	94.9716
SCN-1 Thiocyanate ion; NCS-1 equivalent				
-1	302-04-5	785	C(1)N(1)S(1)	58.0777
SO4-2 Sulfate ion; Sulphate ion				
-2	14808-79-8	1270	O(4)S(1)	96.0576